Worksheet 6-4 & 6-5 B
FTC Part 2 and Graphical Analysis #2
AP Calculus

Name_____

Directions: For $1-4$, find $f'(x)$ for the following.

1. $f(x) = \int_{-1}^{x} (t^2 - 2t + 5)dt$

2. $f(x) = \int_{0}^{3x} (\ln(t^2) - 1)dt$

3. $f(x) = \int_{5}^{x^2} \sin(t^3 - t)\, dt$

4. $f(x) = \int_{e^{2x}}^{-3} \left(\sqrt[3]{t}\right)dt$

5. $g(x)$ is a twice differential function defined by $g(x) = \int_{0}^{x} (t^2 - 3t - 28)dt$

a. Find $g'(2)$.

b. On what interval(s) is $g(x)$ increasing? Give a reason for your answer.

c. At which $x-$value(s) does $g(x)$ have a point of inflection? Give a reason for your answer.

d. $h(x) = g(2x)$. Find $h'(3)$ and $h''(3)$.

Directions: For $6-9$, find evaluate the following expressions.

6. $\frac{d}{dx}\int_{2}^{\sin(x)} (t^2+1)dt =$

7. $\frac{d}{dx}\int_{-1}^{x} (e^t - t^2)dt =$

8. $\frac{d}{dx}\int_{2}^{x^2} (\sin(3t+1))dt =$

9. $\frac{d}{dx}\int_{x}^{3x} \ln(4t)dt =$

---

10. $h(x)$ is a twice differential function defined by $h(x) = 2x^2 + \int_{0}^{2x} (t^2 - 3t - 8)dt$.

a. Find $h'(x)$ and $h''(x)$.

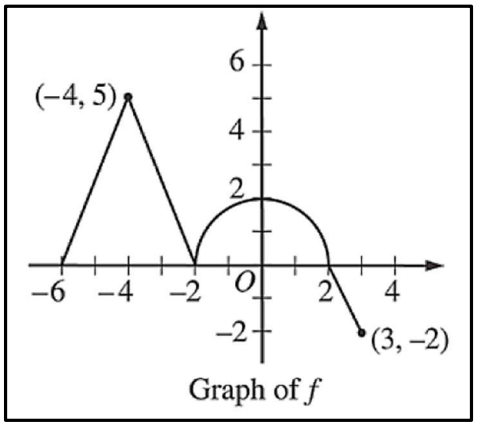
b. Find $\lim_{x\to 0} \frac{h(x)}{x^2+5x}$

c. On what interval(s) is $h(x)$ concave down? Give a reason for your answer.

d. At which x value(s) does $h(x)$ have a relative maximum? Give a reason for your answer.

11.) **From the 2013 IPE (AB5)**

The graph of the continuous function f , consisting of three line segments and a semicircle, is shown at the right. Let g be the function given by $g(x) = \int_{-2}^{x} f(t)dt$.



Graph of f

(a) Find $g(-6)$ and $g(3)$

(b) Find $g'(0)$

(c) Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a horizontal tangent. Determine whether g has a local maximum, a local minimum, or neither at each of these values. Justify your answers.

(d) Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a point of inflection. Explain your reasoning.